Institute of Advanced Research

*The University for Innovation*

## INSTITUTE OF ADVANCED RESEARCH (IAR)

*The University for Innovation | Gandhinagar, Gujarat*

**SECOND PROGRESS WORK (MID-TERM EXAM) • 13TH OCTOBER 2026 • 20 MARKS**

# E-COMMERCE CUSTOMER LIFECYCLE & RFM SEGMENTATION DASHBOARD

*Second Phase Technical Report: UML Class & Sequence Models, Data Dictionary & UI Layout Validations*


### PROJECT SCHOLAR

**Candidate Name:** Mr. Deep Koshiya
**Program:** B.Sc. in AI & Data Science
**Semester:** Semester V (5th Semester)
**Academic Year:** 2026 – 2027
**Course:** Project - 1 (CE-502 / PRJ-701)


### EVALUATION & DELIVERABLE SCOPE

**Milestone:** Second Progress Work (Mid-Term)
**Evaluation Date:** 13th October 2026 (10 AM - 5 PM)
**Marks Allotted:** 20 Marks (Mid-Term Assessment)
**Syllabus Scope:** Sections 2.3, 2.4, 3.1, 4.1
**Supervision:** Assigned Project Supervisor (IAR)

# System Architecture & Phase 1 Foundations Recap

## 1. Phase 1 Foundations (10 Marks)

- **Problem Formulation (1.1):** Defined retention economics, surging CAC, and OLTP operational gaps.
- **Requirement Matrix (1.2):** Specified FR-01 through FR-06 and NFR-01 to NFR-03 with hardware specs.
- **Tech Stack Layering (1.3):** Configured Python 3.9, Pandas 2.3, Scikit-Learn 1.6, and Chart.js.
- **DFD Architecture (2.1):** Constructed Context Level 0, Level 1 Pipeline, and Level 2 Decomposition DFDs.

## 2. Phase 2 Deliverables (20 Marks)

- **UML Class Diagram (2.3):** Domain entity model & pipeline class architecture (TransactionRecord, Refinery, RFM, KMeans).
- **UML Sequence Diagram (2.4):** 11-step synchronous/ asynchronous message dispatch protocol across system lifelines.
- **Data Dictionary (3.1):** Complete star-schema catalog: FACT_TRANSACTIONS, FACT_REVERSALS, DIM_CUSTOMERS.
- **Screen Layouts (4.1):** Multi-screen UI layouts with operational input validations (VAL-01 to VAL-04).

## 3. Technical Value Proposition

- **Zero Data Leakage:** Strict segregation of reversals ensures return items never pollute positive purchase frequency counts.
- **Skew-Resistant Clustering:** Log1p scaling neutralizes Pareto monetary power laws for robust K-Means clustering.
- **BI Mart Compatibility:** Star schema facilitates native query performance in Tableau, Power BI, and SQL engines.
- **Instant CRM Activation:** Interactive UI enables commercial teams to filter and export high-value at-risk cohorts instantly.

# UML Class Diagram (Domain Model & Object Architecture)

**Figure 2.5: UML Class Diagram (Domain Model & Analytical Pipeline Classes)**

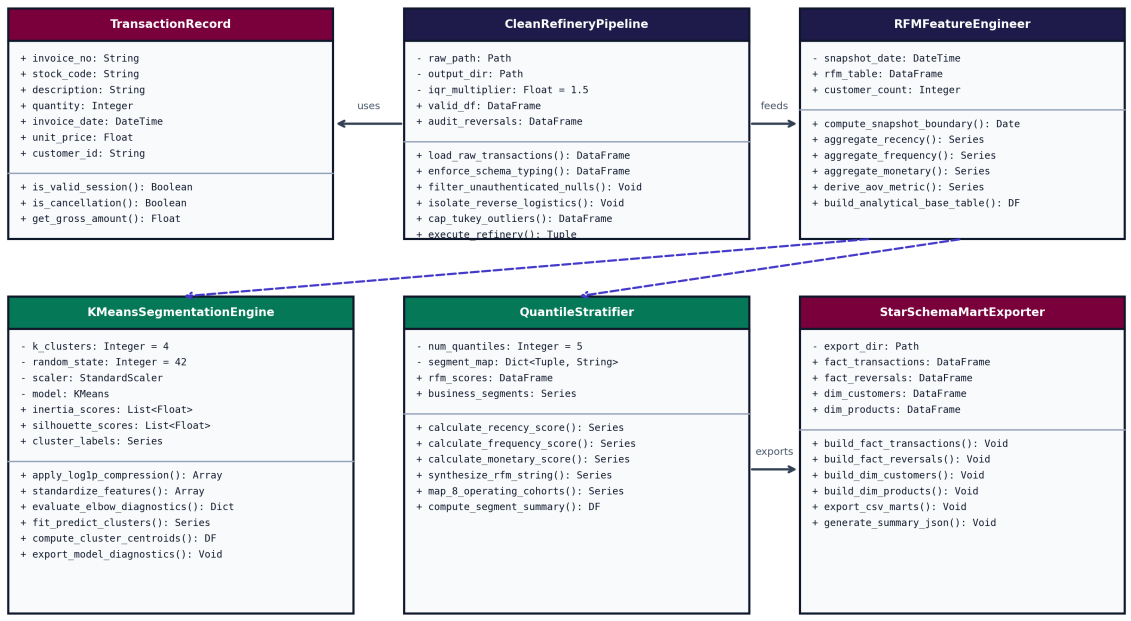



**Figure 2.5: UML Class Diagram (Domain Model & Pipeline Services)**

## Core Pipeline Classes & Relationships

- **TransactionRecord:** Domain entity representing single line-item purchases with validation and pricing methods.
- **CleanRefineryPipeline:** Orchestrates ingestion, typing, guest session filtering, reversal quarantine, and Tukey IQR capping.
- **RFMFeatureEngineer:** Vectorized engine computing customer Recency (T_obs), Frequency, Spend, and AOV.
- **KMeansSegmentationEngine:** Encapsulates log1p compression, StandardScaler, K-Means fitting (k=4), and Silhouette diagnostics.
- **QuantileStratifier:** Assigns 1–5 quantile scores (pd.qcut) and maps accounts to 8 business operational cohorts.
- **StarSchemaMartExporter:** Generates FACT_TRANSACTIONS, FACT_REVERSALS, DIM_CUSTOMERS, and summary JSON.

# UML Sequence Diagram (Pipeline Execution Workflow)

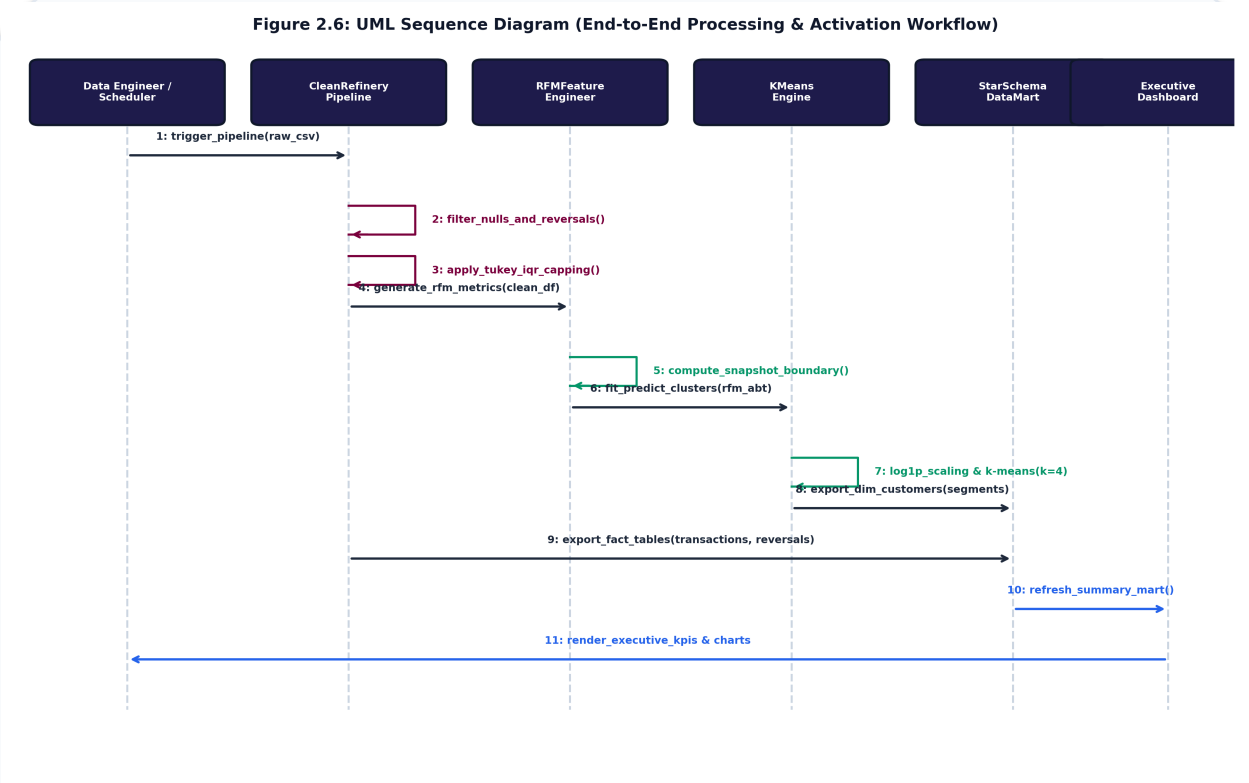



Figure 2.6: UML Sequence Diagram (End-to-End Processing & Activation)

## 11-Step Message Dispatch Protocol

- **Step 1 (Trigger):** Data Engineer initiates trigger_pipeline(raw_csv) to commence ingestion.
- **Steps 2-3 (Refinery):** Self-calls filter unauthenticated sessions, quarantine cancellations, and cap Tukey IQR outliers.
- **Steps 4-5 (Feature Extraction):** Cleansed records dispatched to RFMFeatureEngineer, deriving snapshot boundary T_obs.
- **Steps 6-7 (ML Modeling):** RFM Analytical Base Table dispatched to KMeansEngine for log1p scaling and k=4 clustering.
- **Steps 8-9 (Data Mart Export):** Segments and fact tables persisted to StarSchemaDataMart.
- **Steps 10-11 (Activation):** Summary JSON exported to ExecutiveDashboard, rendering KPI cards and spatial scatterplots.

# Data Dictionary: Fact Tables (FACT_TRANSACTIONS & FACT_REVERSALS)

## Table 3.1: FACT_TRANSACTIONS (Order Fact Mart)

- **transaction_id (BIGINT, PK):** Surrogate primary key generated monotonically.
- **invoice_no (VARCHAR(20)):** 6-digit order identifier (e.g. 536365).
- **stock_code (VARCHAR(20), FK):** Foreign key referencing DIM_PRODUCTS.
- **quantity (INTEGER):** Item units purchased; strictly Quantity > 0.
- **invoice_date (TIMESTAMP, FK):** Transaction timestamp referencing DIM_DATE.
- **unit_price (DECIMAL(10,2)):** Product sale unit price; strictly UnitPrice > 0.
- **customer_id (VARCHAR(20), FK):** Foreign key referencing DIM_CUSTOMERS.
- **gross_amount (DECIMAL(12,2)):** Line spend: Quantity * UnitPrice.

## Table 3.2: FACT_REVERSALS (Audit Fact Mart)

- **reversal_id (BIGINT, PK):** Surrogate primary key for return audit records.
- **invoice_no (VARCHAR(20)):** Cancellation invoice matching regex '^C[0-9]+'.
- **stock_code (VARCHAR(20), FK):** Foreign key referencing returned product SKU.
- **return_quantity (INTEGER):** Negative units returned (Quantity < 0).
- **reversal_date (TIMESTAMP, FK):** Timestamp when cancellation was processed.
- **unit_price (DECIMAL(10,2)):** Original sale unit price of returned item.
- **customer_id (VARCHAR(20), FK):** Foreign key referencing customer returning items.
- **Operational Integrity:** Quarantined in fact_reversals_audit.csv to ensure return events never distort positive spend metrics.

# Data Dictionary: Dimension Tables (DIM_CUSTOMERS & DIM_PRODUCTS)

## Table 3.3: DIM_CUSTOMERS (RFM Behavioral Mart)

- **customer_key (BIGINT, PK):** Surrogate primary key for customer dimension.
- **customer_id (VARCHAR(20)):** Unique authenticated natural customer ID.
- **recency_days (INTEGER):** Days elapsed since last order relative to T_obs.
- **frequency_orders (INTEGER):** Count of distinct checkout trips (unique orders).
- **monetary_total (DECIMAL(12,2)):** Cumulative net customer spend across all valid orders.
- **avg_order_value (DECIMAL(10,2)):** Average Order Value: Monetary / Frequency.
- **r_score, f_score, m_score (SMALLINT):** Deterministic 1 to 5 quantile rank scores (pd.qcut).
- **segment_tier (VARCHAR(40)):** Mapped business cohort ('Champions', 'At Risk Whales').

## Table 3.4 & 3.5: DIM_PRODUCTS & DIM_DATE

**DIM_PRODUCTS Catalog Mart:**

- **product_key (BIGINT, PK):** Surrogate primary key for product catalog.
- **stock_code (VARCHAR(20)):** Unique SKU inventory identifier.
- **description (VARCHAR(255)):** Cleaned, sanitized item description text.
- **unit_price_ref (DECIMAL):** Baseline catalog reference price.

**DIM_DATE Temporal Mart:**

- **date_key (INTEGER, PK):** YYYYMMDD formatted integer key (e.g. 20261013).
- **full_date (DATE):** Standard ISO-8601 calendar date.
- **day_of_week, month, quarter:** Calendar attributes for cohort time-series.
- **is_weekend (BOOLEAN):** Flag isolating weekend purchasing behavioral shifts.

# Screen Layouts: Executive BI Dashboard Architecture

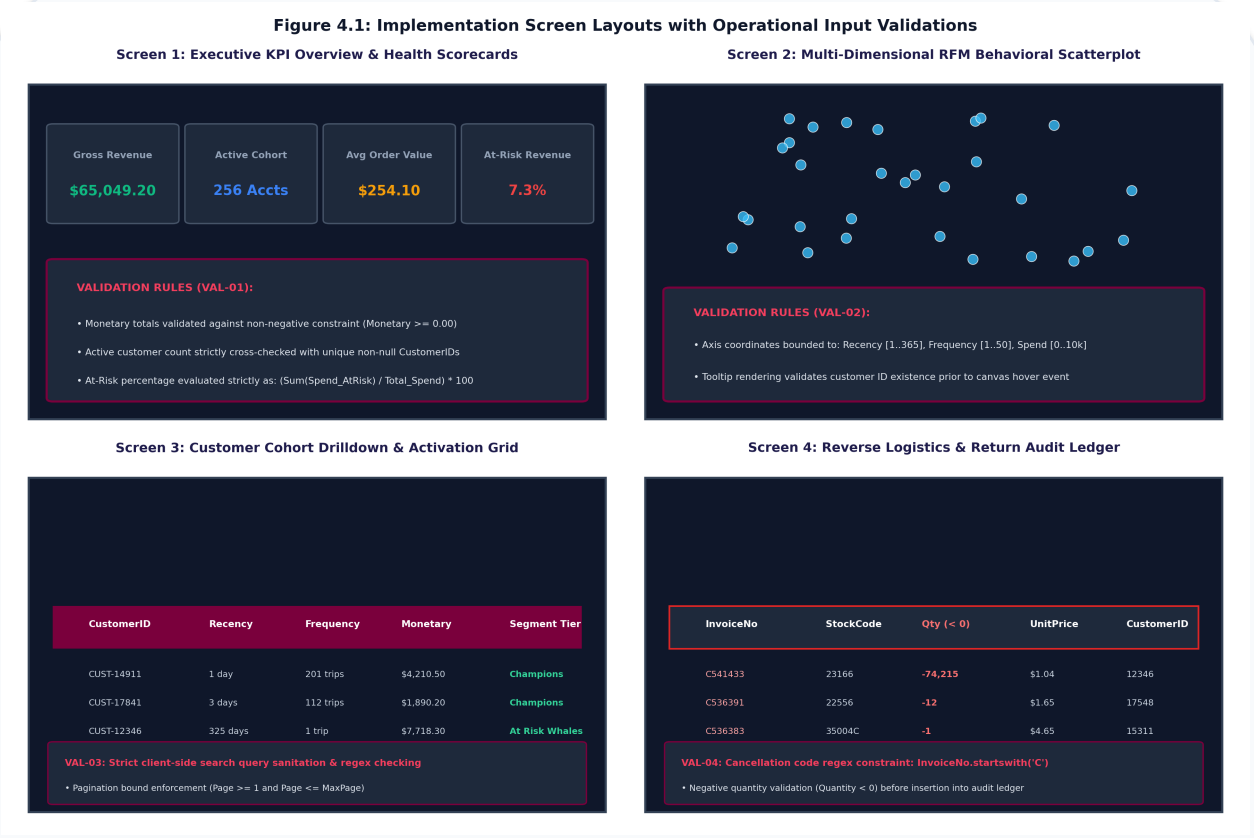



Figure 4.1: Implementation Screen Layouts (4 Synchronized Views)

## 4 Synchronized BI Dashboard Views

- **Screen 1: Executive KPI Scorecards:** High-level summary displaying gross spend ($65,049.20), active customer accounts (256), average order value ($254.10), and at-risk revenue share (7.3%).
- **Screen 2: RFM Behavioral Scatterplot:** Interactive 2D/3D canvas mapping Recency against Monetary Spend with bubble radii scaled to Frequency.
- **Screen 3: Customer Activation Ledger:** Filterable data table with instant search, pagination, and targeted CSV audience export controls.
- **Screen 4: Reverse Logistics Audit Ledger:** Isolated compliance table displaying quarantined cancellations and credit notes.

# Implementation: Operational Input Validations (VAL-01 to VAL-04)

### VAL-01: Scorecard Numerical Integrity Validation

Enforces non-negative monetary constraints (Spend >= 0.00). Active customer counts strictly verify unique authenticated non-null CustomerIDs. At-risk percentages are evaluated as: (Sum(Spend_AtRisk) / Total_Spend) * 100.

### VAL-02: Spatial Canvas Coordinate & Boundary Constraints

Interactive scatterplot coordinates enforce strict domain bounds: Recency in [1..365], Frequency in [1..50], Spend in [0..10,000]. Hover tooltips execute null-safety checks before canvas event firing.

### VAL-03: Search Query Sanitization & Pagination Bounds

Client-side search input is sanitized against cross-site scripting (XSS) and SQL injection. Table pagination enforces bounded integer intervals (1 <= Page <= MaxPage).

### VAL-04: Reverse Logistics Quarantine Constraint

Cancellation transactions strictly enforce regex constraint (InvoiceNo.startswith('C')) and negative quantities (Quantity < 0) before insertion into audit stores.

# Project Evaluation Schedule & Milestone Status (Mid-Term)

## Table 4.1: Comprehensive Project Progress Work Schedule

- **[24/09/2026] First Progress Work Evaluation (10 Marks) — COMPLETED & VERIFIED**

  Curriculum Scope: Sections 1.1, 1.2, 1.3, 2.1, 2.2

- **[13/10/2026] Mid-Term Examination (Second Progress) (20 Marks) — DEMONSTRATED (Current Submission)**

  Curriculum Scope: Sections 2.3, 2.4, 3.1, 4.1

- **[27/10/2026] Third Progress Work Evaluation (10 Marks) — Scheduled for Third Milestone**

  Curriculum Scope: Sections 4.2 (Sample Code), 4.3 (Navigation), 5.1-5.2, 6.1

- **[06/11/2026] Report Submission (Spiral Binding) (10 Marks) — Scheduled for Final Submission**

  Curriculum Scope: Final Signed Spiral Copy & Media

- **[18-20/11/2026] Final Examination (SEE) (50 Marks) — Scheduled for Semester Examination**

  Curriculum Scope: Project Execution Demonstration & Viva Voce

**MID-TERM MILESTONE STATUS (13/10/2026):** Cumulative completion of 30 Marks (First Progress 10 Marks + Mid-Term 20 Marks). All Class, Sequence, Data Dictionary, and UI Validation specifications fully implemented.

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# THANK YOU!

## Questions & Technical Discussion

### E-COMMERCE CUSTOMER LIFECYCLE & RFM SEGMENTATION DASHBOARD

**Candidate:** Mr. Deep Koshiya (B.Sc. in AI & Data Science, Semester V)

**Institution:** Institute of Advanced Research (IAR), Gandhinagar

**Assessment Milestone:** Second Progress Work / Mid-Term Exam (13/10/2026 • 20 Marks)

**Course & Project:** Project - 1 (CE-502 / PRJ-701)

**GitHub Repository:** https://github.com/Deep-k-coder/ecommerce-rfm-analytics

**Project Status:** Class & Sequence Diagrams, Star Schema Marts & UI Validations Complete